

Project Title: Simple File Explorer Application in C++ (Linux OS)

Overview:

The File Explorer Application is a command-line-based file management system developed in **C++** using the **C++17 <filesystem> library**. It enables users to navigate directories and perform various file operations such as creating, deleting, copying, moving, searching, and modifying file permissions directly from the terminal. This project demonstrates practical implementation of file system handling and exception management in modern C++.

Objective:

To design a lightweight and user-friendly file management tool that provides basic file system operations without relying on external GUI libraries, thereby improving understanding of file handling and system programming concepts in C++.

Key Features:

- **List Directory (ls):** Displays the contents of the current directory.
- **Change Directory (cd):** Navigates into subdirectories or back to parent directories.
- **Create File (create):** Creates a new empty file.
- **Delete (delete):** Removes files or directories recursively.
- **Copy (copy):** Copies files or folders with overwrite support.
- **Move (move):** Moves or renames files and folders.
- **Search (search):** Recursively searches for a file by name.
- **Change Permissions (chmod):** Modifies file access permissions using octal format (e.g., 644).
- **Help (help):** Displays the list of available commands and their descriptions.
- **Exit (exit):** Terminates the program gracefully.

Technology Stack:

- **Programming Language:** C++
- **Libraries Used:** <filesystem>, <iostream>, <fstream>, <string>
- **Compiler:** g++ (with C++17 support)
- **Operating System:** Linux / Unix-based OS

Applications:

- Managing files and folders efficiently through a simple command-line interface.
- Understanding file system operations and permissions in Linux.
- Demonstrating practical usage of modern C++ features like the **filesystem API**.

Conclusion:

The Simple File Explorer Application provides a robust and interactive platform for performing essential file operations using C++. It highlights the capabilities of the C++17 filesystem library and

serves as a foundational project for learners exploring system programming, command-line tools, and file management concepts.

```
$ ./file_explorer
===== Simple File Explorer (C++ - LinuxOS) =====
Type 'help' to view available commands.

[/home/kashinath]$ ls

Contents of "/home/kashinath":
[DIR] Documents
[DIR] Downloads
[FILE] notes.txt
[FILE] todo.txt

[/home/kashinath]$ create demo.txt
File created: demo.txt

[/home/kashinath]$ ls

Contents of "/home/kashinath":
[DIR] Documents
[DIR] Downloads
[FILE] notes.txt
[FILE] todo.txt
[FILE] demo.txt

[/home/kashinath]$ copy demo.txt backup.txt
Copied demo.txt to /home/kashinath/backup.txt
```

```
[/home/kashinath]$ move backup.txt demo_renamed.txt  
Moved backup.txt to /home/kashinath/demo_renamed.txt
```

```
[/home/kashinath]$ search demo_renamed.txt  
Searching for "demo_renamed.txt" in "/home/kashinath"...  
Found at: /home/kashinath/demo_renamed.txt
```

```
[/home/kashinath]$ chmod demo_renamed.txt 755  
Permissions updated for: demo_renamed.txt
```

```
[/home/kashinath]$ delete demo_renamed.txt  
Deleted: demo_renamed.txt
```

```
[/home/kashinath]$ help
```

Available commands:

ls	- List files and directories
cd <dir>	- Change directory
create <file>	- Create a file
delete <file/dir>	- Delete file or directory
copy <src> <dest>	- Copy file or folder
move <src> <dest>	- Move or rename file/folder
search <filename>	- Search file recursively
chmod <file> <mode>	- Change file permission (octal)
help	- Show available commands
exit	- Quit program

```
[/home/kashinath]$ exit  
Exiting File Explorer...
```